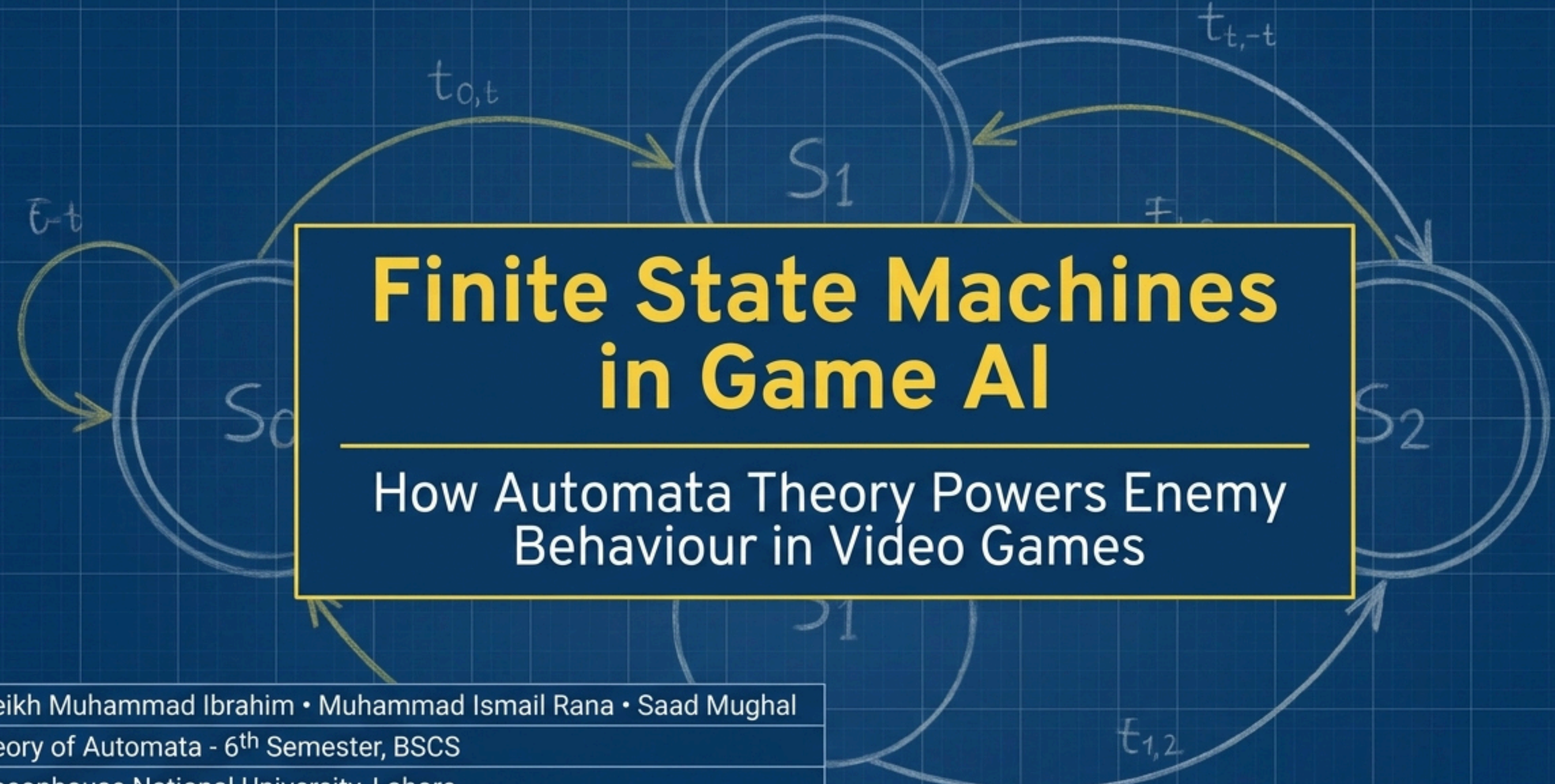




Finite State Machines in Game AI

How Automata Theory Powers Enemy
Behaviour in Video Games

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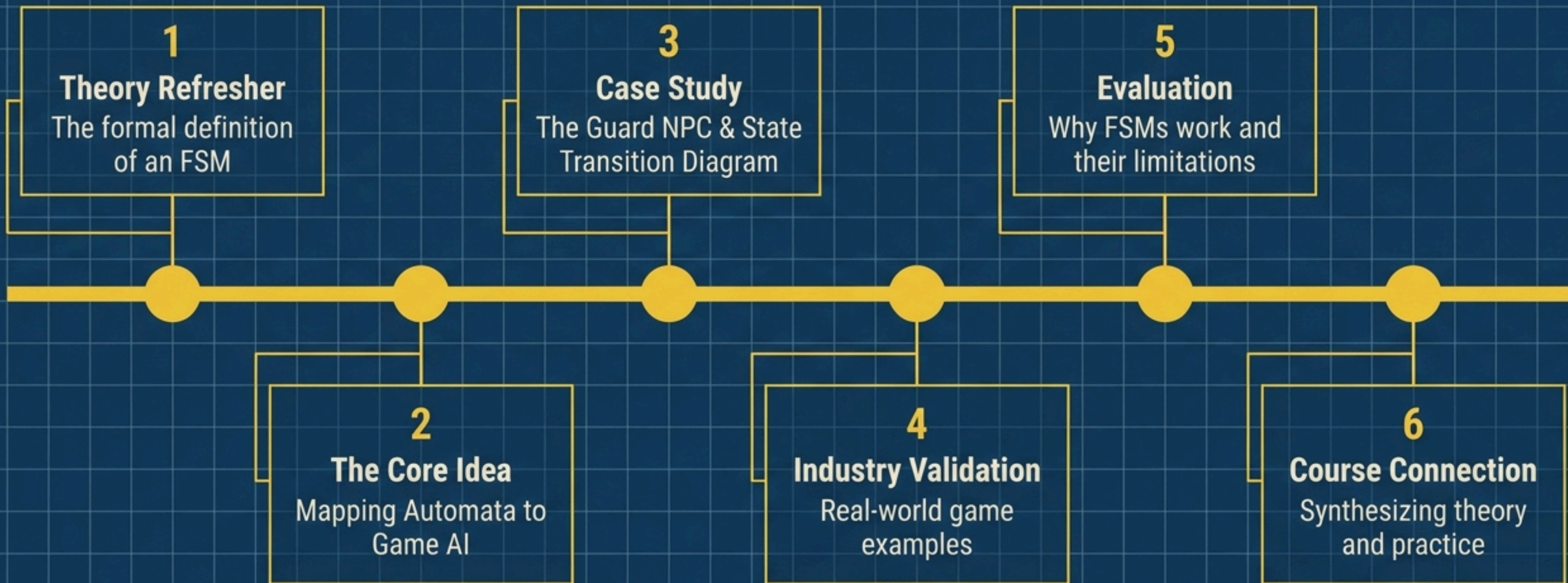
Theory of Automata - 6th Semester, BSCS

Beaconhouse National University, Lahore

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Presentation Outline



The Formal 5-Tuple Definition

Q: A finite set of states
(all the possible conditions
the machine can be in)

Σ : A finite set of inputs
(the things that can
cause a change)

δ : The transition function
(rules: given a state and
an input, what state do
we go to?)

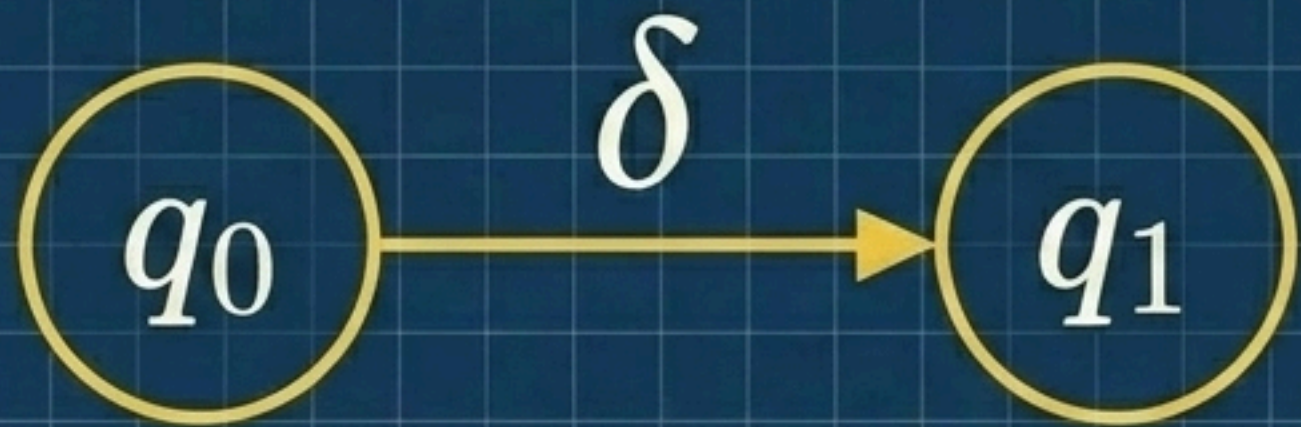
$$M = (Q, \Sigma, \delta, q_0, F)$$

q_0 : The start state
(where the machine begins)

F: The set of accept states
(optional in the game AI context)

Translating Math into Mechanics

Theory



Game developers use FSMs to control Non-Player Characters (NPCs). NPC behaviors (patrolling, chasing) act directly as FSM states.

Game AI



Game Event: See Player
Acts as Input (Σ)

Behavior: Chasing
Acts as State (Q)

Transition functions tell the NPC exactly when to switch behaviors. Designers are essentially building finite automata using game terms.

Applying the Transition Function (δ)

States (Q)



Patrol (q_0)



Chase (q_1)



Attack (q_2)

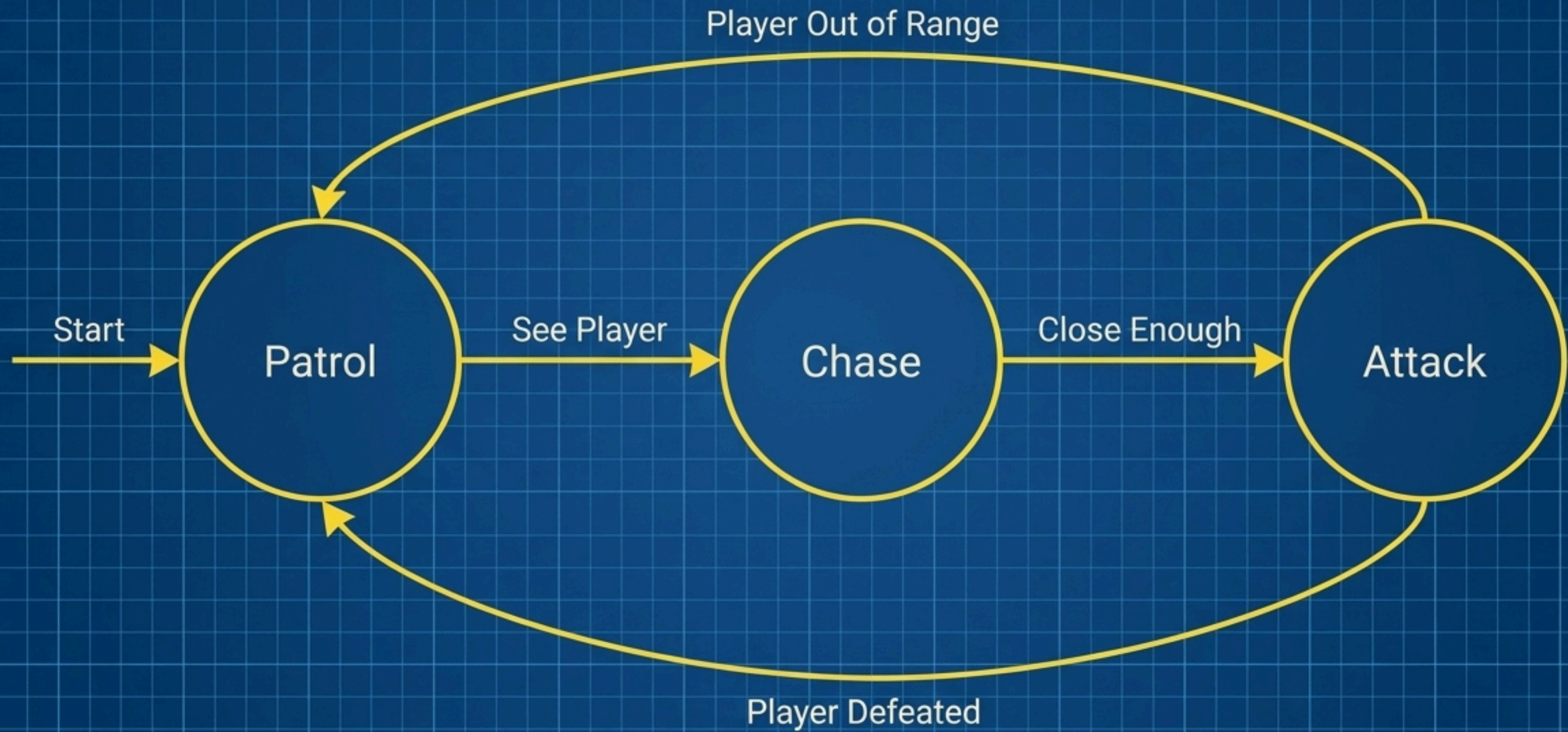
$\delta(\text{Patrol}, \text{See Player}) = \text{Chase}$

$\delta(\text{Chase}, \text{Close Enough}) = \text{Attack}$

$\delta(\text{Chase}, \text{Player Out of Range}) = \text{Patrol}$

$\delta(\text{Attack}, \text{Player Defeated}) = \text{Patrol}$

Visualizing the Guard NPC Logic



Automata Theory in Industry



Pac-Man (1980)

Ghosts switch between Chase, Scatter, and Frightened states.



DOOM (1993)

Monsters switch between idle, alert, and attack based on sight/sound.



The Elder Scrolls

NPCs switch between daily routines and combat states.



Halo

Soldiers coordinate patrolling, taking cover, attacking, and retreating.

4 Key Developer Advantages

1

Simple

Every behaviour is clearly defined with no confusion.

2

Easy to Design

Developers can draw FSM diagrams before writing any code.

3

Predictable

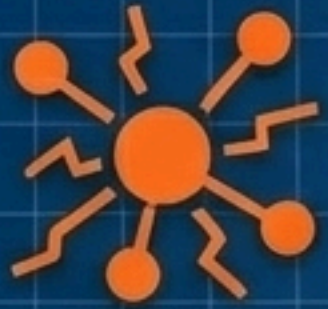
Fixed rules ensure the NPC always behaves the same way given identical inputs.

4

Lightweight

Uses very little memory, allowing hundreds of NPCs at once.

Where the Basic Model Struggles



State Explosion

Too many behaviors create a massive, unmanageable number of states.



Not Flexible

NPCs follow fixed rules and cannot learn or adapt to past player actions.



Hard to Extend

Adding new behaviors requires modifying many existing transitions.

Mapping Automata Theory to Game AI

Course Concept	Game AI Equivalent
Finite set of states (Q)	The NPC's possible behaviors (Patrol, Chase)
Alphabet / Input set (Σ)	Game events (See Player, Close Enough)
Transition function (δ)	The rules deciding when to switch behavior
Start state (q_0)	The NPC's default behavior at the start
Deterministic transitions	Each event always leads to exactly one new state

Key Takeaways

- FSMs are a direct, real-world application of Automata Theory in software.
- States act as behaviors, inputs as events, and transition functions as logic.
- Abstract mathematical models power systems that millions interact with daily.
- Despite advanced AI, FSMs remain foundational due to simplicity and efficiency.

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